DIGITAL SIGNAL ENCODING AND DECODING SIMULATOR

PROJECT SPECIFICATION | DATA COMMUNICATIONS

C11 + C++ GUI

1 PROJECT OVERVIEW

An educational desktop simulator for exploring digital line coding, scrambling, signal visualization, and basic analog-to-digital modulation techniques.

2 FUNCTIONAL FEATURES

DATA GENERATION & ANALYSIS

Random bits, 4/8-zero runs, or manual binary input.

Reports the longest palindrome in the data.

LINE CODING

NRZ-L, NRZ-I, Manchester, Differential Manchester.

Encoding and decoding are available for these methods.

3 ENCODING & OUTPUT

SCRAMBLING

AMI, B8ZS, and HDB3 encoding demonstrations.

MODULATION

Pulse Code Modulation (PCM) and Delta Modulation.

RESULTS

Signal levels, visual plots, and decoded data checks.

4 INPUTS & OUTPUTS

Inputs: binary strings (0/1), encoding method, analog samples, and method-specific parameters (PCM bits/sample; Delta step size).

Outputs: encoded levels, signal display, decoded bits where supported, and a comparison with the original input.

5 TECHNICAL SPECIFICATION

Languages: C11 console application; C++ Windows GUI.

Platform/toolchain: Windows with GCC/MinGW.

Build: compile console sources with GCC; use build_gui.bat for GUI.

Interface: command-line menu and native desktop graphical interface.

6 SCOPE, LIMITS & REQUIREMENTS

- Binary input is limited to 1,000 bits and must contain only 0 and 1.
- Fixed-zero generation supports runs of 4 or 8 zeros.
- Decoding is provided for NRZ-L, NRZ-I, Manchester, and Differential Manchester.
- AMI, B8ZS, and HDB3 are provided as encoding demonstrations.
- PCM and Delta Modulation operate on analog sample values.
- Educational/local simulation only; this project does not transmit physical signals.

PROJECT FILES: C/C++ source, headers, README, and Windows build/run scripts.

Repository: github.com/krishan452006/DataCom-Project